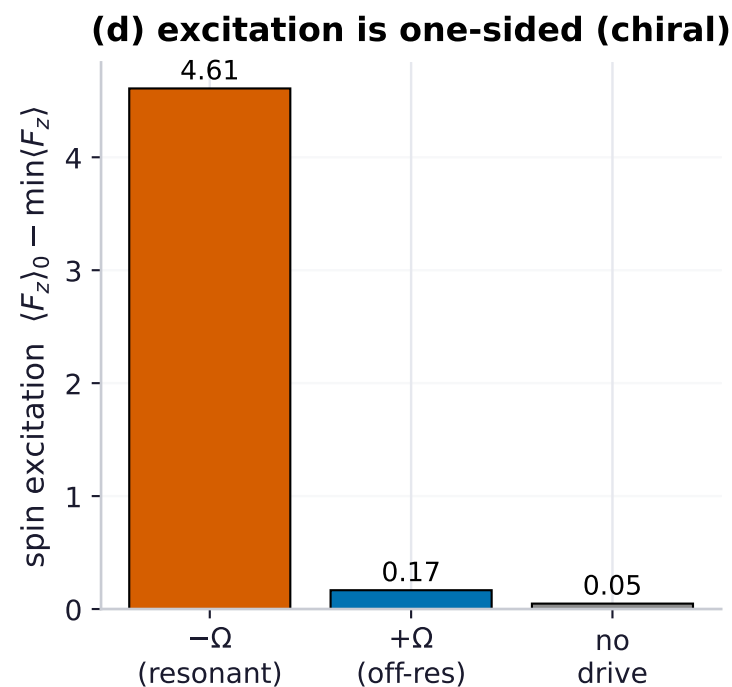
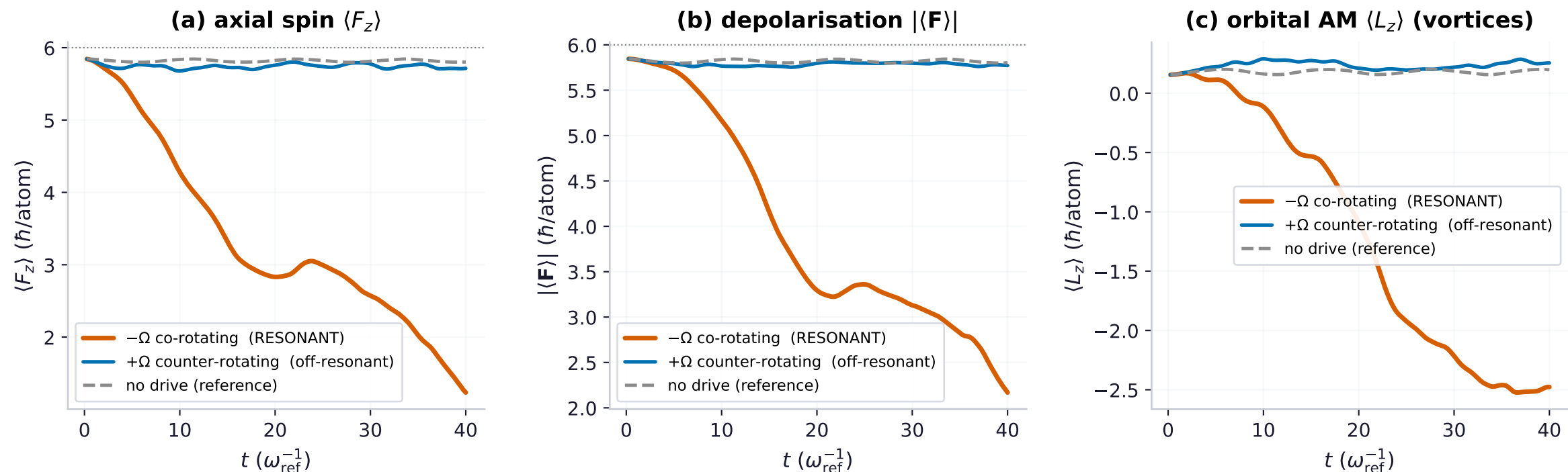
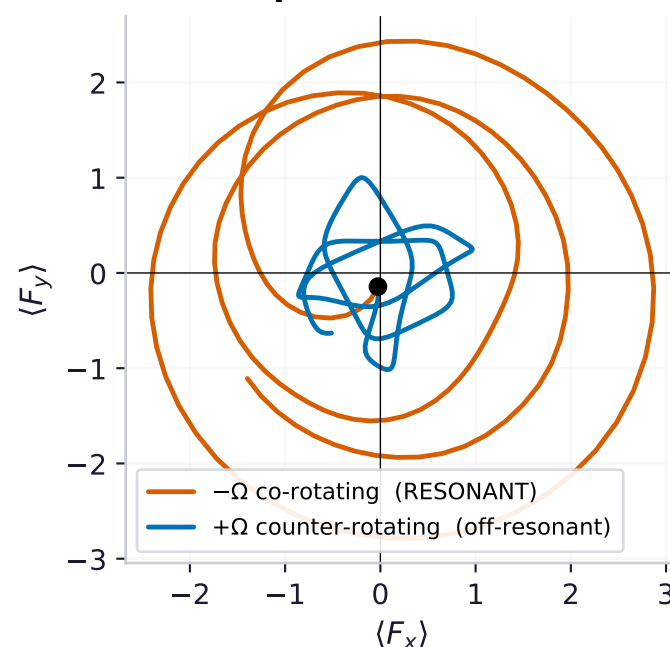


Chiral excitation: co-rotating drive excites, counter-rotating does not (^{151}Eu $F=6$ dipolar BEC, $\Omega = \omega_L=0.5$)



(e) transverse spin: resonant sense spirals in



Magnetic-resonance selection
(top state $m=+6$, static B_z , $\omega_L=0.5$, unitary)

$-\Omega$ CO-rotating (resonant):

$\langle F_z \rangle$: 5.8 $\rightarrow$ 1.2 ($\Delta=4.6$)

$|\mathbf{F}|$: 5.8 $\rightarrow$ 2.2 (depolarised)

peak $|\langle L_z \rangle| = 2.5$ (vortices)

$+\Omega$ COUNTER-rotating (off-res):

$\langle F_z \rangle$: 5.8 $\rightarrow$ 5.7 ($\Delta=0.2$)

$|\mathbf{F}|$: 5.8 $\rightarrow$ 5.8 (unchanged)

peak $|\langle L_z \rangle| = 0.3$ (none)

Selectivity ΔF_z ratio: 28 $\times$

Reverse the field's rotation sense and the
SAME atoms either fully excite or stay frozen.